



FORSTAVIR-EM

Emtricitabine 200mg and Tenofovir Disoproxil Fumarate 300mg Tablets

COMPOSITION:

Each film coated tablet contains

Emtricitabine 200 mg
Tenofovir disoproxil fumarate 300 mg
equivalent to Tenofovir disoproxil 245 mg

DOSAGE FORM

Film coated tablet

PRODUCT DESCRIPTION

FORSTAVIR -EM Emtricitabine 200 mg and Tenofovir Disoproxil Fumarate 300 mg Tablet is White to off-white, modified capsule shaped, film coated tablets, debossed with 'I' on one side '37' on the other side.

Each Tablet contains 200 mg of Emtricitabine & 300 mg of Tenofovir Disoproxil Fumarate which is equivalent to 245 mg of Tenofovir Disoproxil

PHARMACOLOGY

PHARMACODYNAMICS

Emtricitabine is an NRTI that is converted intracellularly to its active metabolite, emtricitabine 5'-triphosphate. Tenofovir DF is an ester prodrug of the NRTI, tenofovir. The prodrug is hydrolysed intracellularly to tenofovir, which is then converted to the active metabolite, tenofovir diphosphate. The active metabolites of the drugs compete with deoxycytidine 5'-triphosphate (emtricitabine 5'-triphosphate) or deoxyadenosine 5'-triphosphate (tenofovir diphosphate) for incorporation into HIV DNA, thereby terminating viral DNA chain growth and inhibiting the activity of the viral reverse transcriptase.

PHARMACOKINETICS

Absorption

The median bioavailability of emtricitabine in fasted patients following a single oral dose is 92%. The median bioavailability of tenofovir in fasted patients following a single oral dose is 25%.

Distribution

Protein binding of emtricitabine is less than 4%; and of tenofovir, is less than 0.7% (plasma) and 7.2% (serum).

Metabolism and Elimination

Emtricitabine undergoes limited metabolism and is primarily excreted by the kidneys. The median terminal elimination half-lives of emtricitabine and tenofovir were 15.5 and 17.6 hours.

INDICATION

Treatment of HIV-1 Infection:

FORSTAVIR-EM is indicated in combination with other antiretroviral agents (such as non nucleoside reverse transcriptase inhibitors or protease inhibitors) for the treatment of HIV-1 infection in adults.

The following points should be considered when initiating therapy with FORSTAVIR-EM for the treatment of HIV-1 infection :

- It is not recommended that FORSTAVIR-EM be used as a component of a triple nucleoside regimen.
- FORSTAVIR-EM should not be coadministered with lamivudine+tenofovir DF+Efavirenz, emtricitabine, tenofovir DF or lamivudine-containing products [see WARNINGS AND PRECAUTIONS].

In treatment experienced patients, the use of FORSTAVIR-EM should be guided by laboratory testing and treatment history.

Pre-exposure prophylaxis (PrEP):

Emtricitabine/Tenofovir Disoproxil is indicated in combination with safer sex practices for pre - exposure prophylaxis to reduce the risk of sexually acquired HIV-1 infection in adults at high risk.

When prescribing FORSTAVIR-EM for PrEP, health care providers must:

- Prescribe FORSTAVIR-EM as part of a comprehensive prevention strategy because emtricitabine / tenofovir disoproxil is not always effective in preventing the acquisition of HIV-1 infection.
- Counsel all uninfected individuals to strictly adhere to the recommended FORSTAVIR-EM dosing schedule because the effectiveness of emtricitabine / tenofovir disoproxil in reducing the risk of acquiring HIV-1 was strongly correlated with adherence as demonstrated by measurable drug levels in clinical trials.
- Confirm a negative HIV-1 test immediately prior to initiating FORSTAVIR-EM for a PrEP indication. If clinical symptoms consistent with acute viral infection are present and recent (<1 month) exposures are suspected, delay starting PrEP for at least one month and reconfirm HIV-1 status, including acute or primary HIV-1 infection; and - screen for HIV-1

infection at least once every 3 months while taking emtricitabine / tenofovir disoproxil for PrEP.

DOSAGE AND METHOD OF ADMINISTRATION

Treatment of HIV-1 infection:

The dosage of FORSTAVIR-EM is one tablet (containing 200 mg of emtricitabine and 300 mg of tenofovir DF) once daily taken orally with or without food.

Dose adjustment for renal impairment

Significantly increased drug exposures occurred when emtricitabine or tenofovir DF were administered to patients with moderate to severe renal impairment. Therefore, the dosing interval of FORSTAVIR-EM should be adjusted in patients with baseline creatinine clearance of 30-49 ml/min using the recommendations in Table 1. The safety and effectiveness of these dosing interval adjustment recommendations have not been clinically evaluated; therefore, clinical response to treatment and renal function should be closely monitored in these patients .

No dose adjustment is necessary for patients with mild renal impairment (creatinine clearance of 50-80 ml/min). Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment [see WARNINGS AND PRECAUTIONS].

Table 1: Dosage Adjustment for Patients with Altered Creatinine Clearance

	Creatinine Clearance (ml/min) ^a		
	≥ 50	30-49	<30 (including patients requiring hemodialysis)
Recommended Dosing Interval	Every 24 hours	Every 48 hours	FORSTAVIR- EM should not be administered

a= Calculated using ideal (lean) body weight

Pre-exposure prophylaxis (PrEP):

This indication is based on clinical trials in men who have sex with men (MSM) at high risk for HIV-1 infection and in heterosexual serodiscordant couples.

Posology

Prevention of HIV in adults: One tablet, once daily.

Adults with renal impairment

Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment. FORSTAVIR-EM should only be used in individuals with creatinine clearance (CrCl) < 80 mL/min if the potential benefits are considered to outweigh the potential risks

Table 2: Dosing recommendations in adults with renal impairment

	Pre-exposure prophylaxis
Mild renal impairment (CrCl 50-80 mL/min)	Limited data from clinical studies support once daily dosing in HIV-1 uninfected individuals with CrCl 60-80 mL/min. Use is not recommended in HIV-1 uninfected individuals with CrCl < 60 mL/min as it has not been studied in this population
Moderate renal impairment (CrCl 30-49 mL/min)	Not recommended for use in this population.
Severe renal impairment (CrCl < 30 mL/min) and haemodialysis patients.	Not recommended for use in this population.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATION

FORSTAVIR-EM is contraindicated in patients with previously demonstrated hypersensitivity to any of the components of the product

WARNING AND PRECAUTIONS

Overall HIV-1 infection prevention strategy

Emtricitabine/Tenofovir Disoproxil is not always effective in preventing the acquisition of HIV-1. The time to onset of protection after commencing emtricitabine/tenofovir disoproxil is unknown.

Emtricitabine/Tenofovir Disoproxil should only be used for pre-exposure prophylaxis as part of an overall HIV-1 infection prevention strategy including the use of other HIV-1 prevention measures (e.g., consistent, and correct condom use, knowledge of HIV-1 status, regular testing for other sexually transmitted infections).

Risk of resistance with undetected HIV-1 infection:

Emtricitabine/Tenofovir Disoproxil should only be used to reduce the risk of acquiring HIV-1 in individuals confirmed to be HIV negative. Individuals should be

re-confirmed to be HIV-negative at frequent intervals (e.g., at least every 3 months) using a combined antigen/antibody test while taking emtricitabine/tenofovir disoproxil for pre-exposure prophylaxis.

Emtricitabine/Tenofovir Disoproxil alone does not constitute a complete regimen for the treatment of HIV-1 and HIV-1 resistance mutations have emerged in individuals with undetected HIV-1 infection who are only taking emtricitabine/tenofovir disoproxil.

If clinical symptoms consistent with acute viral infection are present and recent (< 1 month) exposures to HIV-1 are suspected, use of emtricitabine/tenofovir disoproxil should be delayed for at least one month and HIV-1 status reconfirmed before starting Emtricitabine/Tenofovir Disoproxil for pre-exposure prophylaxis.

Importance of adherence:

The effectiveness of emtricitabine/tenofovir disoproxil in reducing the risk of acquiring HIV-1 is strongly correlated with adherence as demonstrated by measurable drug levels in blood. HIV-1 uninfected individuals should be counselled at frequent intervals to strictly adhere to the recommended Emtricitabine/Tenofovir Disoproxil daily dosing schedule.

Lactic Acidosis /Severe Hepatomegaly with Steatosis

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of nucleoside analogs alone or in combination with other antiretrovirals. Particular caution should be exercised when administering nucleoside analogs to any patient with known risk factors for liver disease; however, cases have also been reported in patients with no known risk factors. Treatment with FORSTAVIR-EM should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or pronounced hepatotoxicity (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

Patients with HIV and HBV Co-infection

The efficacy and safety of tenofovir disoproxil fumarate-emtricitabine has not been thoroughly studied in the treatment of HIV-HBV co-infection, several studies have indicated that discontinuation of tenofovir disoproxil fumarate -emtricitabine in HBV-infected patients may result in severe and acute exacerbation or flare up of the hepatitis. Liver function tests should be monitored for at least several months in HIV/HBV coinfecting patients who are suspended from treatment with tenofovir disoproxil fumarate emtricitabine. Because of the risk of HIV resistance with two drug therapy, if co-infected patients require treatment, other agents such as adefovir or pegylated interferon should be used.

New Onset or Worsening of Renal Impairment

Emtricitabine and tenofovir are principally eliminated by the kidney. Renal impairment, including cases of acute renal failure and Fanconi syndrome (renal tubular injury with severe hypophosphatemia), has been reported with the use of tenofovir DF [see UNDESIRABLE EFFECTS].

It is recommended that creatinine clearance be calculated in all patients prior to initiating therapy and as clinically appropriate during therapy with tenofovir disoproxil fumarate and emtricitabine. Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients at risk for renal impairment. Dosing interval adjustment of FORSTAVIR-EM and close monitoring of renal function are recommended in all patients with creatinine clearance 30-49 ml/min. No safety or efficacy data are available in patients with renal impairment who received tenofovir DF and emtricitabine using these dosing guidelines, so the potential benefit of FORSTAVIR-EM therapy should be assessed against the potential risk of renal toxicity. FORSTAVIR-EM should not be administered to patients with creatine clearance <30 ml/min or patients requiring hemodialysis. FORSTAVIR-EM should be avoided with concurrent or recent use of a nephrotoxic agent.

Decreases in Bone Mineral Density

Subclinical renal phosphate wasting could possibly contribute to a decrease in bone mineral density. Thus, monitoring of bone mineral density should be considered in patients with history of, or risk factors for, pathologic fractures. No study has examined the benefit of calcium and vitamin D supplementation or bisphosphonates in this setting.

Fat Redistribution

Redistribution/accumulation of body fat including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting, breast enlargement and cushingoid appearance " have been observed in patients receiving antiretroviral therapy.

The mechanism and long-term consequences of these events are currently unknown. A causal relationship has not been established.

Immune Reconstitution Syndrome

Immune reconstitution syndrome has been reported in patients treated with combination antiretroviral therapy, including emtricitabine and tenofovir. During the initial phase of combination antiretroviral treatment, patients whose immune system responds may develop an inflammatory response to indolent or residual opportunistic infections (such as *Mycobacterium avium* infection, cytomegalovirus, *Pneumocystis jirovecii* pneumonia (PCP), or tuberculosis), which may necessitate further evaluation and treatment.

Early Virologic Failure

Caution when using triple nucleoside reverse transcriptase inhibitor (NRTI) therapy. Such treatments are generally less effective than triple drug regimens containing 2 NRTIs in combination with either a non nucleoside reverse transcriptase inhibitor (NNRTI) or a protease inhibitor; early virologic

failure and high rates of resistance substitutions have been reported. Patients receiving triple NRTI therapy should be carefully monitored and considered for treatment modification.

INTERACTIONS WITH OTHER MEDICAMENTS

Any medications that reduce renal function may increase concentration of emtricitabine and tenofovir disoproxil fumarate. No drug interactions with clinical consequences are known for emtricitabine. Atazanavir should always be ritonavir boosted (atazanavir 300 mg should be boosted with ritonavir 100 mg) when used with tenofovir disoproxil fumarate.

Tenofovir disoproxil fumarate should be used very cautiously with didanosine due to increased rates of adverse reactions including peripheral neuropathy and pancreatitis. If co-administration of didanosine and tenofovir disoproxil fumarate is necessary, didanosine dose should be adjusted and patients should be closely monitored for didanosine-related adverse reactions. The recommended didanosine dose for patients weighing ≥60 kg in this setting is 250 mg; however, there is no adequate information for patients weighing below 60 kg.

Patients with Renal Impairment

Because both tenofovir disoproxil fumarate-emtricitabine are excreted via the kidneys, modification of tenofovir disoproxil fumarate-emtricitabine dosage and special caution is necessary when treating patients with renal impairments.

PREGNANCY AND LACTATION

Pregnancy Category B:

No adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, FORSTAVIR-EM should be used during pregnancy only if clearly needed.

Lactation

Tenofovir disoproxil fumarate-Emtricitabine is excreted in breast milk and should not be used while nursing. The Centres for Disease Control and Prevention recommend that HIV-infected women do not breast feed their infants to avoid risking postnatal transmission of HIV.

UNDESIRABLE EFFECTS

The most frequently reported adverse reactions considered possibly or probably related to emtricitabine and /or tenofovir disoproxil fumarate were nausea and diarrhoea.

Grade 2 through 4 laboratory abnormalities occurred in patients in the Tenofovir disoproxil fumarate/emtricitabine. Grade 3 and 4 laboratory abnormalities occurring in patients in the tenofovir disoproxil fumarate/emtricitabine included elevated creatine kinase, triglycerides, neutropenia and amylase.

SYMPTOMS AND TREATMENT OF OVERDOSE

If overdose occurs, the patient must be monitored for evidence of toxicity, and standard supportive treatment applied as necessary

EFFECT ON ABILITY TO DRIVE AND USE MACHINE

No studies on the effects on the ability to drive and use machines have been performed. However, patients should be informed that dizziness has been reported during treatment with both emtricitabine and tenofovir disoproxil fumarate.

PRECLINICAL SAFETY DATA

NA

INSTRUCTION FOR USE

It is preferable that Forstavir-EM is taken with food.

STORAGE CONDITION

Do not store above 30°C. Store in the original package in order to protect from moisture. Keep the bottle tightly closed.

SHELF LIFE

24 Months.

THERAPEUTICS CODE

J05AR03 - tenofovir disoproxil and emtricitabine

PRESENTATIONS

Plastic container of 30 tablets

MANUFACTURED BY:



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PRODUCT REGISTRATION HOLDER:

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DATE OF REVISION

30.11.2023